JIXIAN LIU

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APPOINTMENTS

Graduate Research Assistant Aug. 2025 - Present
NetD-Lab, Data Science and AI Institute, Advisor: Enrique Mallada
Johns Hopkins University

RESEARCH INTERESTS

Robotics, Robot Learning, Safety-Critical Control, Reinforcement Learning.

EDUCATION

Johns Hopkins University Baltimore, MD Aug. 2024 - Present
Ph.D. Electrical and Computer Engineering

Xi'an Jiaotong University Xi'an, Shaanxi Sept. 2020 - Jul. 2024
B.S. Physics

HONORS & AWARDS

- CoRL Travel Award 2026
- Amazon AI PhD Fellowship ($155,000) 2026
- ACC Travel Grant 2026
- Mathematical Institute for Data Science Fellowship ($25,000) 2026
- Hong Kong PhD Fellowship (Top Award for HK PhD Applicants) 2024
- Dean's Fellow Award, Boston University (Top Award for BU PhD Applicants) 2024
- MITACS Globalink Research Internship Scholarship (C$7000) 2023
- Top-10 Undergraduate Students of the Year (Top Individual Award in XJTU) 2023
- National Scholarship (Top 1%) 2022, 2023
- Finalist Winner, Mathematical Contest in Modeling (Top 2% around the World) 2022
- First Place in Shaanxi, The Chinese Mathematics Competitions for College Students 2021
- Presidential Scholarship, Xi'an Jiaotong University (Top 5‰) 2021

RESEARCH PUBLICATIONS

Conferences & Workshops

1. **Jixian Liu**, Ihab Tabbara, Hussein Sibai and Enrique Mallada, "On the Limits of Sampling-Based Reachability: Geometry, Dynamics, and Sample Complexity", 10th Annual Conference on Robot Learning, 2026.
2. Zhuo Ouyang, **Jixian Liu**, and Enrique Mallada, "Symplectic Inductive Bias for Data-Driven Target Reachability in Hamiltonian Systems", 65th IEEE Conference on Decision and Control, 2026.
3. **Jixian Liu** and Enrique Mallada, "Safety-Critical Control via Recurrent Tracking Functions", 2026 American Control Conference, 2026.
4. **Jixian Liu** and Enrique Mallada, "Recurrent Control Barrier Functions: A Path Towards Nonparametric Safety Verification", 64th IEEE Conference on Decision and Control, 2025.

5. **Jixian Liu**, Tao Xu, Jianping He, and Chongrong Fang, "Smart Predict-then-Optimize Method with Dependent Data: Risk Bounds and Calibration of Autoregression", arXiv preprint arXiv:2411.12653, 2024.

Journals

1. **Jixian Liu**, Ihab Tabbara, Hussein Sibai and Enrique Mallada, "Fundamental Limits of Sampling-Based Reachability: Geometry and Flow Distortion", arXiv preprint arXiv:2610.40473, 2026.
2. Xueyuan Cui, **Jixian Liu**, and Yao Zhou, "Enabling Net Zero Energy Buildings With Shared Storage: A Cyber-Physical Perspective", IEEE Internet of Things Journal, 2023.

Datasets & Benchmarks

1. Yiyou Sun, Xinyang Han, Weichen Zhang, ..., **Jixian Liu**, Maojia Song, Ruheng Wang, Shengcao Cao, Yanxin Zhang, et al., "Agents' Last Exam," arXiv preprint arXiv:2606.05405, 2026.

ACADEMIC SERVICES

Reviewer of Journals

- IET Renewable Power Generation Journal
- IEEE Control Systems Letters

Reviewer of Conferences & Workshops

- American Control Conference 2026, 2027
- IEEE Conference on Decision and Control 2026
- China Automation Congress 2024

TEACHING

Teaching Assistant

- Foundation of Reinforcement Learning (Online), Johns Hopkins University Fall 2025, Fall 2026
- Control System Design, Johns Hopkins University Fall 2025

ADVISING AND MENTORING

Mentoring

Zhuo Ouyang Undergrad Student, Peking University

College of Engineering Next position: CS Ph.D. at UIUC (Advisors: Tong Zhang and Nan Jiang)

PRESENTATIONS

Talks

- Oct. 2026, "On the Limits of Sampling-Based Reachability: Geometry, Dynamics, and Sample Complexity," Invited talk, hosted by Lars Lindemann, Zurich, Switzerland
- May 2026, "Safety-Critical Control via Recurrent Tracking Functions", American Control Conference, New Orleans, LA, USA
- Dec. 2025, "Recurrent Control Barrier Functions: A Path Towards Nonparametric Safety Verification", 64th IEEE Conference on Decision and Control, Rio de Janeiro, RJ, Brazil
- Oct. 2025, "Recurrence is All You Need: From Parametric to Nonparametric Safety", Institute for Assured Autonomy Graduate Student Lightning Talks, Baltimore, MD, USA

Poster

- Nov 2026 "On the Limits of Sampling-Based Reachability: Geometry, Dynamics, and Sample Complexity", 10th Conference on Robot Learning, Austin, TX, USA

- Oct 2026, "On the Limits of Sampling-Based Reachability: Geometry, Dynamics, and Sample Complexity", JHU & Amazon AI2AI Research Festival, Baltimore, MD, USA
- May 2025, "Recurrent Control Barrier Functions: A Path Towards Nonparametric Safety Verification", 2nd Northeast Systems and Control Workshop, NYC, USA

REFERENCES

Enrique Mallada Ph.D. Advisor
Associate Professor, Department of Electrical and Computer Engineering
Johns Hopkins University, Baltimore, MD

Jianping He
Professor, School of Automation and Intelligent Sensing
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Xiaoming Duan
Associate Professor, School of Automation and Intelligent Sensing
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